

Elevated TCDD in chicken eggs and farm-raised catfish fed a diet with ball clay from a Southern United States mine.

The U.S. Food and Drug Administration (FDA) terminated the use of ball clay from a mine in Mississippi as an additive in animal feed after discovering nanogram per gram concentrations of 2,3,7,8-tetrachlorodibenzo-p-dioxin (2,3,7,8-TCDD). The FDA collected chicken eggs and farm-raised catfish in affected areas and throughout the remaining continental United States to assess levels of 2,3,7,8-TCDD. A new method using quadrupole ion storage tandem-in-time mass spectrometry (QISTMS) measured the 2,3,7,8-TCDD levels in 42 catfish fillet composites, 3 Tilapia fillet composites, 46 chicken egg samples, and 6 chicken feeds. Six catfish composites and 20 egg samples had 2,3,7,8-TCDD concentrations significantly above 1.0 pg/g wet weight of fillet or whole egg. Farm-raised catfish not exposed to feed containing ball clay had a mean 2,3,7,8-TCDD concentration of 0.12 pg/g. The TCDD isomer pattern in ball clay differed from the TCDD isomer pattern in a fly ash sample and from the "chick edema factor" TCDD pattern in a sample of reference toxic fat used as a feed ingredient in the 1950s.